

STEADY THREE-DIMENSIONAL FLOW TO A WELL IN A TWO-LAYERED AQUIFER

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Abstract--In Part I, a solution is obtained for the potential distribution induced by a well screened in uniform sand and pumping from it at a constant rate. This sand is overlain by another sand of different permeability, over whose upper surface the potential is uniform. The potential remains constant also at the common outer boundary of both sands, which is concentric with the well. A sample calculation is made for an assumed set of the parameters of the system. For all practical values of the parameters in leaky aquifers, the validity of earlier assumptions made to reduce three-dimensional problems to two dimensions is verified.

In Part II, the solution obtained is for the case where the steadily discharging well is screened in the upper of the two sands. In this case, the two layers of sand are bounded above and beneath by impervious strata. The other boundary conditions are the same as those stipulated for the problem described in Part I. From the general solution, the potential distribution is obtained for the flow towards a well partially penetrating a single sand.

PART I--WELL TAPPING LEAKY AQUIFER SEMI-CONFINED ABOVE

Introduction--By assuming the leakage through an aquiclude into a confined aquifer to be vertically downward and totally refracted at their interface, and by considering the leaking water to be uniformly distributed throughout the thickness of the aquifer, JACOB [1946] reduced the three-dimensional radial flow toward a well in a leaky aquifer to a two-dimensional one and obtained the following steady-state solution

$$s = (Q/2\pi T) [K_0(r/B) - K_0(r_e/B) I_0(r/B)/I_0(r_e/B)] \dots\dots\dots (1)$$

where Q is the discharge of the well, T is the transmissibility of the confined aquifer, I_0 and K_0 are respectively the modified zero-order Bessel functions of the first and second kinds, r_e is the radius of the outer boundary, and $B = \sqrt{Kbb'/K'}$, b and b' being the thicknesses of the bottom and top beds, and K and K' their respective hydraulic conductivities, with $K \gg K'$.

Eq. (1) is valid only when the above restrictions are met. The present paper develops the general steady-state solution with these restrictions removed.

In Figure 1 are shown the two horizontal beds of different hydraulic conductivities, K' in the upper bed, and K in the lower. The two beds are saturated with water from the impervious base

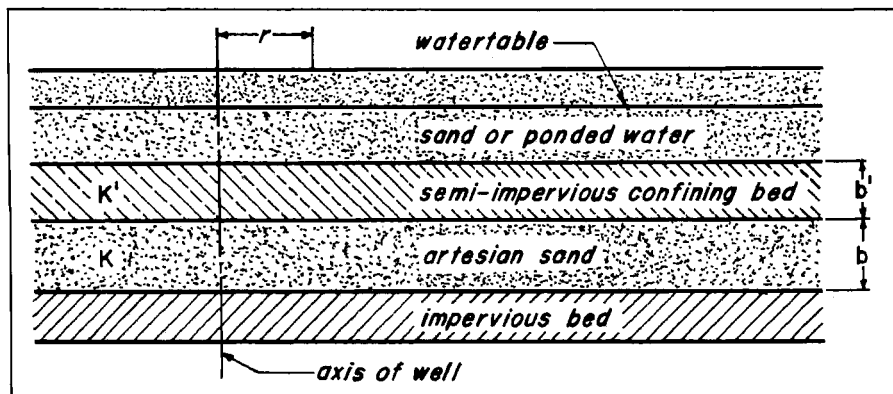


Fig. 1--Diagrammatic representation of a stratified artesian aquifer

of the lower bed to the top of the upper bed, which is covered with ponded water or overlain by a highly transmissive sand with a water table. The upper bed is of thickness b' and the lower is of thickness b . A well, idealized as a sink, is screened in the lower bed and is located at the center of the circular aquifer, around whose outer boundary the potential is maintained at a constant uniform value. If the well is pumped at a constant rate, the induced flow within the region is radially symmetrical and three dimensional in nature. The drawdowns in the upper and lower beds are designated by s' and s respectively.

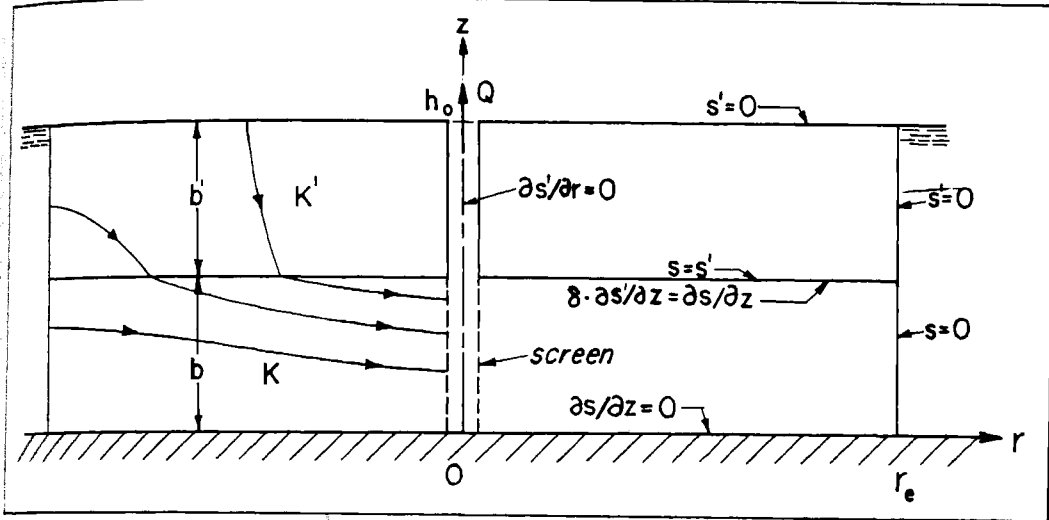


Fig. 2--Diagrammatic representation of steady three-dimensional flow to a well in a two-layered aquifer

Development of the solution--The system of flow being radially symmetrical, the natural coordinates will be those of the cylindrical system, with the origin as shown in Figure 2. The Laplace equation for ground water flow is then written as

$$\nabla^2 s = (\partial^2 s / \partial r^2) + (1/r) (\partial s / \partial r) + (\partial^2 s / \partial z^2) = 0$$

where $s = s(r, z)$ is the drawdown at a point whose coordinates are r and z . Analytically the problem may be further stated to be that of finding simultaneous solutions of the following system of equations

$$\left. \begin{aligned} \nabla^2 s' &= 0 && \text{in the upper bed} \\ s'(r, h_0) &= 0 && \text{where } h_0 = b + b' \\ \partial s'(0, z) / \partial r &= 0 \\ s'(r_e, z) &= 0 \end{aligned} \right\} \dots \dots \dots (2)$$

$$\left. \begin{aligned} \nabla^2 s &= 0 && \text{in the lower bed} \\ \partial s(r, 0) / \partial z &= 0 \\ \lim_{r \rightarrow 0} \left\{ 2 \pi K r \int_0^b [\partial s(r, z) / \partial r] \partial z \right\} &= -Q \\ s(r_e, z) &= 0 \end{aligned} \right\} \dots \dots \dots (3)$$

$$s'(r, b) = s(r, b) \dots \dots \dots (4)$$

$$\delta [\partial s'(r, b) / \partial z] = \partial s(r, b) / \partial z \dots \dots \dots (5)$$

where $\delta = (K'/K)$.

It is easily shown that Conditions (2) and (3) are satisfied by the following solutions

$$s' = (Q/2\pi T) \sum_{n=1}^{\infty} A_n J_0(\alpha_n r) \sinh[\alpha_n (h_0 - z)] \dots\dots\dots (6)$$

$$s = (Q/2\pi T) [-\ln(r/r_e) + \sum_{n=1}^{\infty} B_n J_0(\alpha_n r) \cosh(\alpha_n z)] \dots\dots\dots (7)$$

where $T = Kb$ and where the α_n are such that $J_0(\alpha_n r_e) = 0$; that is, the $(\alpha_n r_e)$ are zeroes of the Bessel function of the first kind of zero order.

Setting $z = b$ in (6) and (7), it is clear that if Conditions (4) and (5) are to be satisfied, the A_n and B_n must satisfy the equations

$$\sum_{n=1}^{\infty} J_0(\alpha_n r) [A_n \sinh(\alpha_n b') - B_n \cosh(\alpha_n b)] = -\ln(r/r_e) \dots\dots\dots (8)$$

$$\sum_{n=1}^{\infty} J_0(\alpha_n r) [\delta A_n \cosh(\alpha_n b') + B_n \sinh(\alpha_n b)] = 0 \dots\dots\dots (9)$$

Eq. (8) and (9) are Fourier-Bessel series. The coefficients of these series are determined by the usual method in the interval (0 to r_e). These are

$$\delta A_n \cosh(\alpha_n b') + B_n \sinh(\alpha_n b) = 0 \dots\dots\dots (10)$$

$$A_n \sinh(\alpha_n b') - B_n \cosh(\alpha_n b) = 2/[\alpha_n r_e J_1(\alpha_n r_e)]^2 \dots\dots\dots (11)$$

where J_1 is the Bessel function of the first kind of first order.

Solving (10) and (11) for A_n and B_n , and inserting them in (6) and (7), then making the substitution $x_n = \alpha_n r_e$, one obtains

$$s'/(Q/2\pi T) = 4 \sum_{n=1}^{\infty} \frac{\sinh(x_n b/r_e) \sinh[x_n (h_0 - z)/r_e] J_0(x_n r/r_e)}{[x_n J_1(x_n)]^2 \left\{ (1 + \delta) \cosh[x_n (b + b')/r_e] - (1 - \delta) \cosh[x_n (b - b')/r_e] \right\}} \dots\dots\dots (12)$$

$$s/(Q/2\pi T) = -\ln(r/r_e) - 4 \sum_{n=1}^{\infty} \frac{\delta \cosh(x_n b'/r_e) \cosh(x_n z/r_e) J_0(x_n r/r_e)}{[x_n J_1(x_n)]^2 \left\{ (1 + \delta) \cosh[x_n (b + b')/r_e] - (1 - \delta) \cosh[x_n (b - b')/r_e] \right\}} \dots\dots\dots (13)$$

These equations describe the drawdown in the system. They are rapidly convergent and lend themselves to relatively easy calculation.

It is to be observed that if $\delta = 0$, leakage does not take place. Then, as should be expected, s becomes $[-(Q/2\pi T) \ln(r/r_e)]$.

Comparison with approximate solution--In practical flow-systems, δ is often quite small and the thicknesses of the beds are small compared to their radial dimensions. In such cases where δ is small and $h_0 \ll r_e$, (12) and (13) may be approximated if it is remembered that $\sinh(x) = x$ for small values of x . Thus

$$s'/(Q/4\pi T) = [4b(h_0 - z)/\delta r_e^2] \sum_{n=1}^{\infty} R_n \dots\dots\dots (14)$$

$$s/(Q/4\pi T) = -2 \ln(r/r_e) - 4 \sum_{n=1}^{\infty} (R_n/x_n^2) \dots \dots \dots (15)$$

where $R_n = [J_0(x_n r/r_e)]/[(x_n B/r_e)^2 + 1] J_1^2(x_n)$ and where, as defined before, $B = \sqrt{bb'/\delta}$.

That the leakage into the artesian sand is proportional to the drawdown is evident from the following analysis. First, (14) can be put in the form

$$s'/(h_0 - z) = (4b/\delta r_e^2) (Q/4\pi T) \sum R_n$$

Then

$$\partial[s'/(h_0 - z)]/\partial z = [1/(h_0 - z)] (\partial s'/\partial z) - s'/(h_0 - z)^2 = 0$$

or

$$(\partial s'/\partial z) = -s'/(h_0 - z) \dots \dots \dots (16)$$

Using (16) with the relation

$$V_z = -K' (\partial s'/\partial z)$$

one obtains

$$V_z = K's'/(h_0 - z) \dots \dots \dots (17)$$

At the interface of the two beds, $z = b$ and $s = s'$. Making these substitutions, (17) becomes

$$V_z = K's/b' \dots \dots \dots (18)$$

which is the relation used in the approximate theory. In this case, z is taken positively upward.

Eq. (15) clearly indicates that for small δ the flow is almost purely radial and independent of the third dimension, as was assumed by JACOB [1946]. Drawdown values obtained by (15) are in close conformity with those obtained from the approximate theory of (1). Table 1 gives drawdown values computed by the two equations for the same set of parameters used by Jacob.

Table 1--Drawdown values for $T = Kb = 20,000 \text{ ft}^2/\text{day}$,
 $r_e = 100,000 \text{ ft}$, and $B = \sqrt{bb'/\delta} = 20,000 \text{ ft}$

Equations	$s/(Q/4\pi T)$ values for r/ft equal to					
	1000	2000	5000	10,000	20,000	50,000
	and r/r_e equal to					
	0.01	0.02	0.05	0.10	0.20	0.50
(1)	6.228	4.854	3.083	1.849	0.842	0.124
(15)	6.237	4.859	3.081	1.849	0.841	0.123

PART II--WELL TAPPING UPPER OF TWO SANDS WITH NO LEAKAGE

Analysis--In Figure 3 are shown two horizontal beds of different hydraulic conductivities (permeabilities), K in the upper bed, and K' in the lower. The upper and lower beds are respectively of thicknesses b and b' . The two beds are saturated with water from the impervious base of the lower bed to the impervious top of the upper bed. At the outer boundary of the circular aquifer the potential is maintained at a constant uniform value. A well is screened in the upper bed and is located at the center of the circular aquifer. If the well is pumped at a constant rate, the induced flow within the region is radially symmetrical and three dimensional in nature. The drawdowns developed in the upper and lower beds are designated by s and s' respectively.

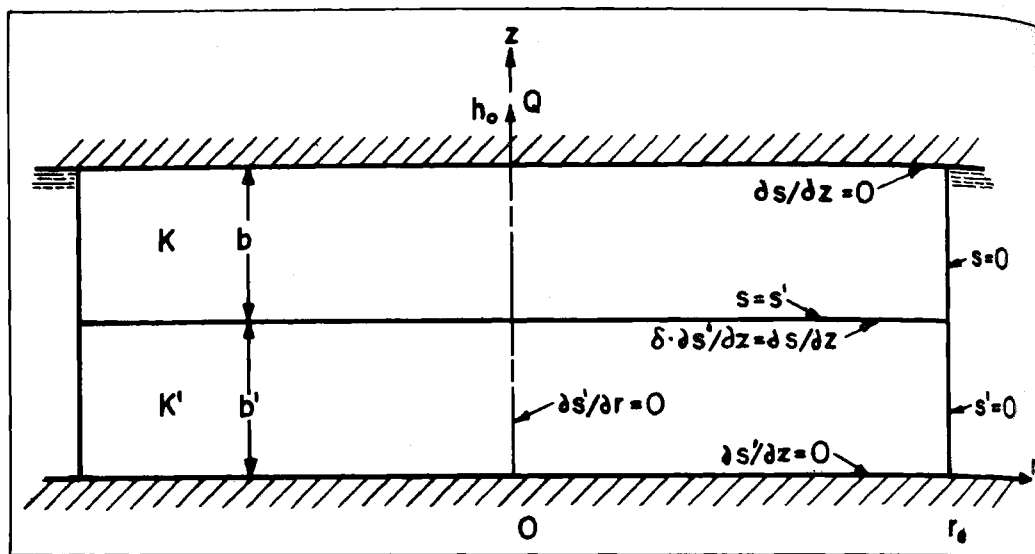


Fig. 3--Diagrammatic representation of a well idealized as a sink tapping upper of two sands of different hydraulic conductivities

Case 1--Vanishing well radius--Because the radius r_w of the well is small compared to the other dimensions of practical interest, the well, for analytical purposes, can be considered as a line sink placed at the axis of the well. The required solution will then be the solutions of the following system of equations (see Fig. 3).

$$\left. \begin{aligned} \nabla^2 s &= 0 \\ s(r_e, z) &= 0 \\ \partial s(r, h_0)/\partial z &= 0 \quad \text{where } h_0 = b' + b \\ \lim_{r \rightarrow 0} \left\{ r \int_{b'}^{h_0} [\partial s(r, z)/\partial r] dz \right\} &= -Q/2\pi K \end{aligned} \right\} \dots \dots \dots (19)$$

$$\left. \begin{aligned} \nabla^2 s' &= 0 \\ s'(r_e, z) &= 0 \\ \partial s'(r, 0)/\partial z &= 0 \\ \partial s'(0, z)/\partial r &= 0 \end{aligned} \right\} \dots \dots \dots (20)$$

$$\partial s(r, b')/\partial z = \delta \cdot \partial s'(r, b')/\partial z \dots \dots \dots (21)$$

$$s(r, b') = s'(r, b') \dots \dots \dots (22)$$

where $\delta = K'/K$, and where $\nabla^2 s = 0$ is the Laplace equation for purely radial ground-water flow.

It is easily shown that Conditions (19) and (20) are satisfied by the following solutions

$$s = (Q/2\pi T) \left[-\ln(r/r_e) + \sum_{n=1}^{\infty} A_n J_0(\alpha_n r) \cosh \alpha_n (h_0 - z) \right] \dots \dots \dots (23)$$

$$s' = (Q/2\pi T) \sum_{n=1}^{\infty} B_n J_0(\alpha_n r) \cosh(\alpha_n z) \dots \dots \dots (24)$$

where $T = Kb$ is the transmissibility of the upper bed, and where the α_n are such that $J_0(\alpha_n r_e) = 0$, that is, the $(\alpha_n r_e)$ are zeroes of the Bessel function of the first kind of zero order. Q is the discharge of the well. A_n and B_n can be found in a manner similar to that followed in Part I. The final solution is given by

$$s/(Q/2\pi T) = \ln(r_e/r) - 4\delta \sum_{n=1}^{\infty} C_n \sinh(x_n b'/r_e) \cosh[x_n(h_0 - z)/r_e] J_0(x_n r/r_e) \dots \dots \dots (25)$$

$$s'/(Q/2\pi T) = 4 \sum_{n=1}^{\infty} C_n \sinh(x_n b/r_e) \cosh(x_n z/r_e) J_0(x_n r/r_e) \dots \dots \dots (26)$$

where $x_n = \alpha_n r_e$, and

$$C_n = 1/[x_n J_1(x_n)]^2 \left\{ (\delta + 1) \sinh(x_n h_0/r_e) + (\delta - 1) \sinh[x_n(b' - b)/r_e] \right\}$$

Eq. (25) and (26) give the full description of the drawdown distribution in the system. They are rapidly convergent and lend themselves to easy calculation.

It is to be observed that if $K' = 0$, or if $\delta = 0$, flow from the lower bed does not occur. Then, as should be expected, the drawdown becomes simply $s = (Q/2\pi T) \ln(r_e/r)$.

In most practical problems, the thickness of the aquifer h_0 is very small compared to its lateral extent, that is, $h_0 \ll r_e$, in which case $\sinh(y) = y$ and $\cosh(y) = 1$. If this approximation is made, the solution can be reduced to a simpler form.

Case 2--Finite well radius--An exact solution for this case (Fig. 4) seems to be difficult to obtain. However, since the radius of the well is very small compared to other dimensions, the assumption that there is no flow into the cylindrical core, of radius r_w , below the well, will result in a very insignificant alteration of the flow pattern.

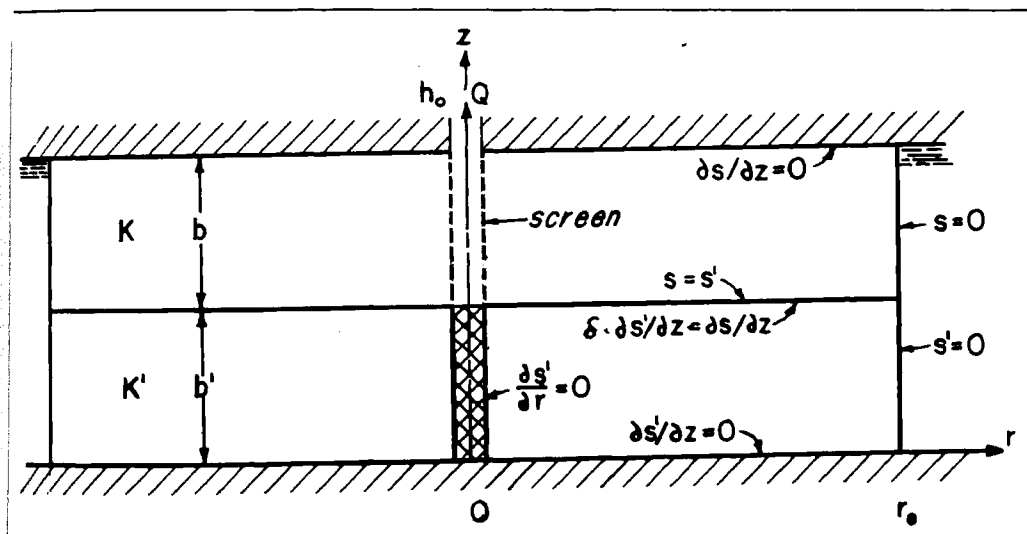


Fig. 4--Diagrammatic representation of a well of finite radius tapping upper of two sands of different hydraulic conductivities

By making this assumption the boundary-value problem will be that given by (19) to (22) if the fourth condition in each of (19) and (20) are replaced respectively by

$$r_w \int_{b'}^{h_0} [\partial s(r_w, z)/\partial r] dz = -Q/2\pi K$$

$$\partial s'(r_w, z)/\partial r = 0$$
$$s/(Q/2\pi T) = \ln(r_e/r) + \delta \cdot \sum_{n=1}^{\infty} F_n \sinh(\beta_n b^*) \cosh \beta_n (h_0 - z) U_0(\beta_n r) \dots \dots \dots$$
$$s'/(Q/2\pi T) = \sum_{n=1}^{\infty} F_n \sinh(\beta_n b) \cosh(\beta_n z) U_0(\beta_n r) \dots\dots\dots$$
$$U_0(\beta_{nr}) = Y_0(\beta_{nr_e}) J_0(\beta_{nr}) - Y_0(\beta_{nr}) J_0(\beta_{nr_e})$$
$$Y_0(\beta_{nr_e}) J_1(\beta_{nr_w}) - Y_1(\beta_{nr_w}) J_0(\beta_{nr_e}) = 0$$
$$F_n = [4J_1(\beta_n r_w) J_0(\beta_n r_e)] / \beta_n r_w [J_1^2(\beta_n r_w) - J_0^2(\beta_n r_e)] \cdot [(\delta + 1) \sinh(\beta_n h_0) + (\delta - 1) \sinh \beta_n y].$$

Partially penetrating wells--If δ is made equal to unity, the solutions obtained above will reduce to those of a well partially penetrating a single aquifer of uniform permeability.

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